

Draft - expert review pending

This unit is complete and passes the platform's automated checks, but no reviewer has read it yet. Treat its sources and claims with care.

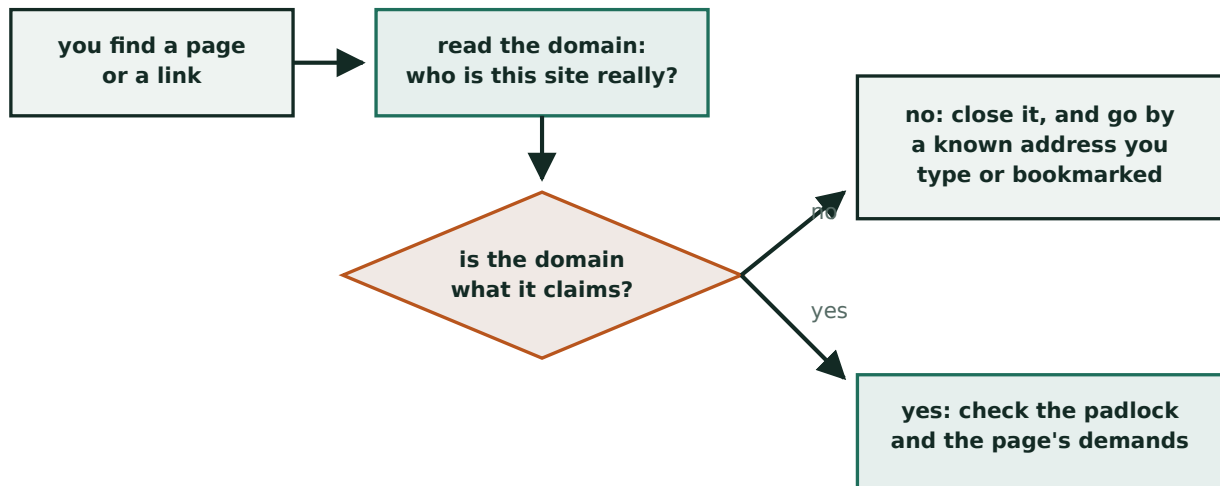
Topic 6.2 - Browsing safely

A real classroom situation

During fee week, a clerk in a Larkana school receives an SMS: "Your school fee account needs verification. Update now: fee-portal-pk.com/verify". The real portal is at a government address. The fake address contains the words "fee" and "portal" and "pk" - all the right words, in the wrong order, in the wrong place. The clerk, who has learned to read addresses, notices that the domain (the part just before the first single slash) is not the government's, closes the message, and opens the real portal by typing its address himself. Nothing happened. That is what safe browsing looks like from the outside: nothing happens, because the trap was seen before it was stepped on.

Evaluating a website before trusting it

The domain is the fact; design is free



Never type a password into a page you did not reach yourself.
The padlock means the connection is encrypted - not that the site is honest.

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Explanation

Safe browsing practices

Safe browsing is Unit 4's habits applied at the moment of the tap, plus one new skill: reading an address. A web address has parts, and the part that identifies the site is the domain - the words just before the first single slash. In "fee-portal-pk.com/verify", the domain is "fee-portal-pk.com": everything before it (such as "gov.pk.") and after the slash is decoration an attacker can choose freely. The clerk's check was one question: is the domain the organisation's real one?

Around that skill sit the checks that catch most traps. **Read the domain before the slash**, not the words in the link text. **Look for the padlock** - it means the connection to the site is encrypted, so nobody in between can read it; it does NOT mean the site is honest, because fake sites can have padlocks too. **Judge the content**: does the page ask for a password, a code, or money? Real institutions rarely demand all three in a hurry. **Prefer the known address**: for anything involving accounts or payments, type the address yourself or use your own bookmark rather than a link from a message - which is exactly the phishing defence from Unit 4. And **keep the browser updated**,

because its warnings (the red screens that announce dangerous sites) are maintained by the update.

Five checks - and what each one cannot tell you

Check	What it tells you	What it does NOT tell you
reading the domain	who really owns the site	nothing about how pretty the page looks
the padlock	the connection is encrypted	whether the site itself is honest
judging the content	whether the page's demands are plausible	nothing about the design quality
using a known address	you are at the real site	what you choose to type there
browser warnings	the site is known to be dangerous	that an unlisted site is safe

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One more habit completes the set: scepticism toward perfection. A dangerous page often looks better than the real one - clean design, urgent official tone, all the right logos. Design is free; the domain is not. When the two disagree, believe the domain.

And for pupils, the teaching version of all this is one sentence, repeated: never type a password into a page you did not reach yourself. The clerk's story, the head teacher's story from Unit 4, and every fake-portal case in between reduce to that sentence.

Activity: Address reading drill

Work in pairs. About 15 minutes.

1. Your teacher shows eight addresses (some real, some constructed), such as "hec.gov.pk/scholarships", "hec-scholarships-pk.com/apply", "pk.gov.verify-fee.net", and five others.

2. For each, write the domain (the part before the first single slash) and your verdict: which organisation owns it, and is the page likely to be what it claims?
3. Mark the three addresses a phishing message would most love, and write what makes each tempting.
4. Swap sheets. Argue any disagreement by pointing at the domain, not the words.

Check your understanding

1. Name the part of a web address that identifies the site, and how to find it.
(Remember)
2. Explain what the padlock does and does not tell you. (Understand)
3. The address "rewards.easypaisa-verify-pk.com/claim" arrives by SMS.
Analyse it: who owns the domain, and what should you do? (Apply)
4. State the one sentence you would teach pupils about passwords and pages.
(Remember)

Summary

Safe browsing is one skill plus four habits: read the domain (the words before the first single slash - the only part an attacker cannot fake cheaply), look for the padlock (encrypted connection, not honesty), judge the content's demands, and reach account-and-money pages by a known address you type or bookmark yourself. Design is free and dangerous pages look better than real ones; the domain is the fact. For pupils, one sentence carries it: never type a password into a page you did not reach yourself.

Self-assessment checklist

Sign in to sync your ticks across devices.

- ☐ I can read a domain out of any address and name its owner.
- ☐ I can explain what the padlock does and does not mean.
- ☐ I can analyse a suspicious address and state the safe action.

Try this at your practicum school

Ask the office for the addresses of the three portals the school uses most (fee, results, attendance). Write them down exactly, and check how staff reach them today: typed, bookmarked, or via links in messages. Note one improvement.

Summative task

Design the front of a "read the address" card for the staff room: how to find the domain (with one worked example), the four checks in order, and the two sentences that defeat most fake-portal messages. Use one real-sounding constructed example and mark its parts.

Marking guide (10 marks):

- Finding the domain explained correctly with a worked example: 3 marks.
- The four checks in order, each with its limitation stated (Analyse): 4 marks.
- The two sentences are the right two: 2 marks.
- Fits a card: 1 mark.

Further reading

- National Cybersecurity Alliance. *Stay Safe Online*. <https://staysafeonline.org/>
- Ministry of Information Technology and Telecommunication. (2021). *National Cyber Security Policy*. Government of Pakistan. <https://moitt.gov.pk/>
- Bourgeois, D. T., Smith, J. L., Wang, S., & Mortati, J. (2019). *Information Systems for Business and Beyond* (2nd ed.), chapter 6. Open Textbook Library. <https://open.umn.edu/opentextbooks/textbooks/information-systems-for-business-and-beyond>

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